

Project Name: Quantum Computing for Combinatorial Assignment Optimization

Mentors: Iván Barrientos Salas

Approximate number of Interns: 3

Skill Prerequisite: Solid programming skills in Python. Familiarity with basic quantum computing concepts (superposition, entanglement, gates) and linear algebra. Experience with quantum programming frameworks (e.g., Qiskit, PennyLane). Understanding of combinatorial optimization problems, QUBO formulations, and basic data modeling is highly desirable.

Approximate Workload (# 8-hour days in a week): 3

Brief Description

Complex assignment problems—where multiple agents must be matched with specific resources or tasks under multidimensional constraints—are ubiquitous in enterprise operations, logistics, and scheduling. This project aims to model generalized bipartite matching as a Quadratic Unconstrained Binary Optimization (QUBO) problem and explore its resolution using quantum algorithms. Recognizing the limitations of current NISQ hardware, the project proposes a hybrid architecture: generating affinity scores (cost matrices) through classical or simulated models, and subsequently executing the Quantum Approximate Optimization Algorithm (QAOA) on real quantum hardware to find optimal global assignments, comparing the performance against classical solvers.

Detailed Research Plan

Overview

This project focuses on implementing and analyzing quantum optimization algorithms tailored for complex combinatorial assignment problems. The interns will study the formulation of QUBO models from generic bipartite datasets, prototype them using quantum simulators, and ultimately benchmark the optimization phase using available real quantum hardware. Interns will gain hands-on experience in quantum algorithm design, noise mitigation strategies for NISQ devices, and practical implementation.

Six-Week Work Plan

Week 1: Background Study & Conceptual Foundation

Objective: Understand generalized assignment constraints and quantum optimization foundations.

Activities:

- Review literature on QAOA and QUBO formulations.

- Define the mathematical constraints for a 1-to-1 and 1-to-N matching scenario.
- Set up the development environment with Python and Qiskit.

Week 2: Data Modeling and QUBO Formulation

Objective: Translate the classical matching problem into a quantum-native format.

Activities:

- Create a synthetic dataset representing two disjoint sets (e.g., agents and resources) with randomized affinity scores.
- Develop the mathematical model in QUBO format representing the constraints (e.g., no resource can be assigned twice) and cost functions.

Week 3: Initial Classical Benchmarking & Quantum Implementation

Objective: Establish classical baselines and begin quantum circuit design.

Activities:

- Implement reference classical models (e.g., simulated annealing, greedy algorithms, or exact solvers).
- Develop the initial QAOA circuits for the formulated QUBO using simulators.

Week 4: Hardware Execution and Simulation

Objective: Transition from exact simulators to real quantum hardware environments.

Activities:

- Execute tests on quantum hardware (e.g., IBM Quantum systems).
- Collect initial metrics on solution quality, execution time, and the impact of hardware noise on circuit depth.

Week 5: Error Analysis, Scaling, and Tuning

Objective: Optimize parameters and analyze performance bottlenecks.

Activities:

- Perform hyperparameter tuning (e.g., adjusting QAOA depth p).
- Compare resource overhead versus accuracy across the classical and quantum models.
- Evaluate scaling behavior with incrementally larger bipartite graphs.

Week 6: Documentation and Final Report

Objective: Finalize the project and prepare deliverables.

Activities:

- Write the detailed project report documenting the comparative analysis.
- Prepare the final presentation for the QIntern mini-workshop.
- Open-source the code repository with documentation.

Guidance and Supervision Plan

The internship will follow a structured approach involving weekly group meetings to discuss progress, challenges, and next steps. As the mentor, Iván Barrientos Salas—Software Architect, CEO of San Ángel Digital and Qubit.mx, and Physics student at UNAM specializing in Quantum Computing—will provide direct technical feedback on mathematical formulation, code implementation, and scientific writing. Additional support sessions will be scheduled to bridge the gap between classical enterprise software architecture and quantum algorithm design.

Deliverables

- A well-documented open-source repository containing the synthetic dataset generator, the QUBO formulations, and the QAOA implementation.
- A final technical report detailing the comparative performance between the classical exact solvers and the quantum optimization approach on NISQ hardware.

Plans for Future Work

- Integration of the quantum optimization module into custom enterprise ERP systems to manage dynamic resource allocation.
- Extension of the model to include dynamic variables (e.g., changing resource costs or real-time availability constraints).